

# EvoSensory

## MINI MANUAL · QUICK START & REFERENCE

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Multi-sensory rhythmic timing training, built by a pediatric OT. This guide covers everything you need to run your first session, understand the data, and document it.

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- Why Rhythmic Timing Training Works — The Research
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## THE EVIDENCE

## Why Rhythmic Timing Training Works

Rhythmic timing training has a substantial evidence base built across more than three decades of research in neuroscience, occupational therapy, physical therapy, and special education. The brain's internal timing system — involving the cerebellum, basal ganglia, supplementary motor area, and prefrontal cortex — can be trained through structured, repetitive practice synchronizing movement to an external beat.

When this system is less efficient, the effects are visible in daily life: impulsivity, bilateral coordination difficulty, slow processing speed, reading challenges, handwriting struggles, and poor motor sequencing. EvoSensory measures timing errors to the millisecond and makes them documentable in a way a therapist watching cannot achieve with the naked eye.

| Area                              | Research Finding                                                                                                                                                                                         | EvoSensory Connection                                                                                                                   |
|-----------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------|
| ADHD & Attention                  | Children with ADHD show measurable timing deficits independent of inattention. Structured rhythmic training reduces impulsivity and improves sustained attention through cerebellar-prefrontal pathways. | Early-tap tendency + session trend (drifting earlier) map directly onto ADHD timing profiles.                                           |
| DCD & Bilateral Coordination      | Rhythmic timing training improves alternating limb timing and interlimb synchronization in DCD. Improvements transfer to functional bilateral tasks.                                                     | Late-tap tendency is the characteristic DCD pattern. Levels 1–2 directly train bilateral alternation.                                   |
| Reading & Phonological Processing | Temporal processing underlies phoneme discrimination. Children with dyslexia show significant timing deficits. Rhythmic training improves reading fluency.                                               | Cognitive module — vocabulary spelled by syllable on the beat — directly targets phonological timing.                                   |
| Motor Planning & Praxis           | Sequenced rhythmic training enhances motor program formation and execution speed in praxis deficits.                                                                                                     | Levels 3–5 train increasingly complex sequences. Session trend tracks sequencing under sustained demand.                                |
| TBI & Acquired Injury             | Post-TBI timing deficits correlate with processing speed and cognitive-motor outcomes. Rhythmic training shows measurable improvement.                                                                   | Session trend drifting late may indicate cognitive fatigue during the session itself — useful diagnostic data.                          |
| Parkinson's Disease               | Rhythmic Auditory Stimulation (RAS) is one of the most supported non-pharmacological PD interventions — documented improvements in gait cadence and motor timing.                                        | Spatial audio + wired headphones provides the auditory entrainment mechanism central to RAS. Motion module adds gait movement training. |
| Stroke Rehabilitation             | Rhythmic cuing improves motor timing and bilateral retraining in post-stroke upper and lower extremity rehabilitation.                                                                                   | Motion module (clapping, marching) and unilateral modality settings support targeted stroke rehab protocols.                            |

| Area        | Research Finding                                                                                                                                                     | EvoSensory Connection                                                                                                                                |
|-------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------|
| Handwriting | Graphomotor timing is one of the strongest predictors of handwriting legibility and speed. Timed rhythmic practice produces transfer effects to handwriting fluency. | Classic levels 1–3 build bilateral and timing foundations that transfer to handwriting. Deviation data directly reflects graphomotor timing quality. |

**Why millisecond precision matters**

8ms early versus 62ms early is invisible to the human eye but clinically significant. The 8ms patient has an established internal clock. The 62ms patient has a systematic anticipatory pattern that mirrors impulsivity in daily functioning. EvoSensory makes that distinction visible and documentable in every session.

## STEP BY STEP

## Quick Start

- 1 Open the app**  
Go to [www.evolve-ot.com/evosensory.html](http://www.evolve-ot.com/evosensory.html) in Chrome. Bookmark it or tap Share → Add to Home Screen for easy access next session.
- 2 Create a patient profile**  
Tap **+ New Player**. Enter name or initials, select age category (Pediatric, Adult, or Older Adult), and choose training modality. Starting BPM is set automatically. On phones and tablets, most children tap with one hand — select the single-hand option if that applies.
- 3 Choose a module and level**  
Select **Classic** → **Level 1**. Adjust BPM with + / −. Use Classic with every new patient.
- 4 Confirm hardware**  
A setup screen appears before each session. Check what's connected — wired headphones, device vibration, XAC cushion. Only what you confirm is documented in the SOAP note.
- 5 Tap with the beat**  
Tap the large button, tap the canvas, or press **spacebar** on desktop. One tap per beat. On phone or tablet, tap anywhere on the screen.
- 6 Add observation and generate SOAP note**  
When the session ends, fill in the **Session Observation** field (patient engagement, behavior, fatigue — this populates the S section automatically), then tap **Generate SOAP Note**. Edit before pasting into your EMR.

### Use wired headphones — required for spatial audio

The spatial audio feedback is the core clinical mechanism — early taps produce a tone in the left ear, late taps produce a tone in the right ear. Without wired headphones this feedback is not available. Bluetooth adds 100–200ms lag that defeats this entirely. A \$10 wired earbud is all you need.

## SETUP

## Patient Profiles

| Category          | Hands BPM | Feet BPM | Notes                                  |
|-------------------|-----------|----------|----------------------------------------|
| Pediatric (0–17)  | 60 BPM    | 52 BPM   | Standard starting point                |
| Adult (18–64)     | 66 BPM    | 56 BPM   | Slightly faster baseline               |
| Older Adult (65+) | 52 BPM    | 44 BPM   | Wider hit window applied automatically |

Feet start 8–10 BPM slower: hip → knee → ankle → floor is a longer motor sequence than a hand tap. On phones/tablets, verify bilateral hand use or select the appropriate single-hand modality.

## THE FIVE MODULES

## Choosing the Right Module

| Module                                                                                         | What It Trains                                                                                                        | BPM                             | Requires           |
|------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------|---------------------------------|--------------------|
|  Classic      | Core timing, bilateral coordination, spatial audio + haptic feedback, 6 progressive levels. Start every patient here. | Adjustable                      | Any device         |
|  Oculomotor   | Visual targets move in sync with the beat — pursuits, saccades, depth pulse, gaze hold. Tap-based motor proxy.        | Adjustable /<br>Default: 44 BPM | Any device         |
|  Cognitive    | Vocabulary and math spoken on the beat — phonological processing and number fluency.                                  | Adjustable                      | Any device         |
|  Motion       | Camera detects clapping, knee taps, marching, and jumps timed to the beat.                                            | Adjustable                      | Chrome on computer |
|  Active Games | Body movement controls 5 games — catch stars, pop balloons, kick soccer, duck/jump, balance zones.                    | —                               | Chrome on computer |

### Oculomotor default BPM is 44

Visual tracking needs more time per beat than hand tapping — the eyes must locate the target, track it, and reset before the next beat. Start at 44 BPM and increase gradually.

### Classic — Level Progression

| Level             | Pattern                            | Advance When                                  |
|-------------------|------------------------------------|-----------------------------------------------|
| 1 — Warm Up       | Single button, any key, every beat | 85%+ accuracy, 2 sessions                     |
| 2 — Finding It    | Alternating L/R                    | 85%+ accuracy, 2 sessions                     |
| 3 — In the Groove | Alternating with rests             | 80%+ accuracy, stable trend                   |
| 4 — Steady Beat   | Complex L/R sequences + rests      | 80%+ accuracy, <50ms deviation                |
| 5 — Beat Master   | Irregular patterns + rests         | 80%+ accuracy, consistent timing              |
| 6 — Foot Tap      | Bilateral foot patterns            | Clinical ceiling — maintain for consolidation |

**BPM progression rule:** Increase tempo by 4–8 BPM at a time. If accuracy falls below 50%, reduce BPM before changing the level.

## Motion — Sub-Modes

Chrome on computer required

- Clapping (bilateral UE, waist-up)
- Knee Taps / Midline Crossing (waist-head)
- Marching (full body)
- Jumps (full body + clearance)

## Active Games — Sub-Modes

Chrome on computer required

- Catch Stars (lateral stepping, vestibular)
- Pop Balloons (overhead reach)
- Kick Soccer (unilateral foot)
- Duck & Jump (motor planning)
- Side Steps & Balance (static/dynamic)

## READING THE DATA

## Understanding Your Results

### 1. Accuracy — Were they hitting the beat?

% of beats hit within the acceptable time window. Reflects engagement, attention, and task level. **High accuracy alone does not mean good timing.**

### 2. Timing Deviation — Where exactly did taps land?

Measured in milliseconds (ms). Lower = better. Zero = on the beat. **(+) = early (before the beat)** **(-) = late (after the beat)**

#### Accuracy and deviation are not the same thing

A patient can score 95% accuracy but have +55ms average deviation — hitting the window every time, but always early. That is a separate, clinically meaningful finding that accuracy alone does not capture.

### 3. Session Trend — Did timing change during the session?

Tracks whether errors grew, shrank, or held steady. Drifting earlier may indicate difficulty sustaining inhibitory control. Drifting later may indicate attention fatigue.

## CLINICAL REFERENCE

## Clinical Interpretation Scale

### Accuracy

| Range   | Interpretation                             | Action                                 |
|---------|--------------------------------------------|----------------------------------------|
| 85–100% | <span>Functional</span> Skill established  | Advance level or increase BPM by 4–8   |
| 70–84%  | <span>Emerging</span> Moderate consistency | Continue; target 85% across 2 sessions |
| 50–69%  | <span>Below threshold</span>               | Adjust supports; consider slower BPM   |
| <50%    | <span>Well below threshold</span>          | Reduce BPM by 8–10 or drop one level   |

### Timing Deviation (lower is better)

| Range    | Interpretation                      | Clinical Relevance                                            |
|----------|-------------------------------------|---------------------------------------------------------------|
| 0–20ms   | <span>Excellent</span>              | Strong internal clock; ready for more complex tasks           |
| 21–40ms  | <span>Good</span>                   | Typical for children in active development                    |
| 41–70ms  | <span>Moderate difficulty</span>    | Likely affecting handwriting, bilateral tasks, reading rhythm |
| 71–100ms | <span>Significant difficulty</span> | Document functional impact; direct intervention warranted     |

| Range  | Interpretation    | Clinical Relevance                                   |
|--------|-------------------|------------------------------------------------------|
| 100ms+ | Severe difficulty | Reduce demands; prioritize sensory feedback supports |

Direction & Session Trend

| Pattern                  | May Indicate                                                                      |
|--------------------------|-----------------------------------------------------------------------------------|
| Consistently + (early)   | Impulsivity, response inhibition — common in ADHD and anxiety                     |
| Consistently – (late)    | Processing speed delay, motor initiation difficulty — common in DCD, TBI, fatigue |
| Stable across session    | Consistent regulation throughout                                                  |
| Drifting early over time | Mounting impulsivity or inhibitory control difficulty                             |
| Drifting late over time  | Attention fatigue or processing slowdown under demand                             |

**Key documentation point**  
Always report both accuracy AND deviation. Connect the deviation direction to a functional impact in daily occupations — handwriting, turn-taking, dressing, gait, reading rhythm.



## DOCUMENTATION

## SOAP Note Guide

When the session ends, fill in the **Session Observation** field on the results screen — then tap **Generate SOAP Note**. The note is pre-populated with all session data.

### Session Observation field — fills the S section automatically

Type your clinical observation in the text box on the results screen — patient engagement, behavior, fatigue, comments made during the session. Whatever you enter populates the S section automatically. If left blank, the S section shows a standard fill-in prompt.

| Section | Content                                                                                                     |
|---------|-------------------------------------------------------------------------------------------------------------|
| S       | Populated from Session Observation field — or blank prompt if not filled in                                 |
| O       | Accuracy %, deviation with direction, session trend, level, BPM, modality, age category, hardware confirmed |
| A       | Plain-language interpretation with functional connection prompts and progression status                     |
| P       | Specific next-session BPM and level recommendation based on this session's data                             |

### Always edit before use

Draft only. The A section has fill-in prompts — review and personalize before pasting into your EMR or submitting for billing.

### Insurance documentation tip

Add the functional connection in the A section — how does this patient's timing deficit show up in handwriting, dressing, reading, turn-taking, or gait? The objective numbers are already there.

## CUSTOMIZATION

## Settings Reference

| Setting           | What It Does                                                                                                    |
|-------------------|-----------------------------------------------------------------------------------------------------------------|
| Beat Sound        | Single Beat · Double Beat · 1–2 Spoken (says "one" and "two" — good for young children)                         |
| Audio Feedback    | Spatial audio on/off — left ear tone = tapping early, right ear tone = tapping late. Requires wired headphones. |
| Metronome         | Toggles the beat click — turn off to test internal timing without auditory cue                                  |
| Device Haptics    | Phone or tablet vibration on each tap                                                                           |
| XAC Pillow Rumble | Drives vibration cushion via XAC. Choose <b>Correct tap</b> (reward) or <b>Timing error</b> (error correction). |

| Setting              | What It Does                                                      |
|----------------------|-------------------------------------------------------------------|
| Blind Training Mode  | Fades visuals after 10s — trains audio/haptic timing only         |
| Therapist View       | Shows ms deviation and clinical notes in real time during session |
| Session Setup Prompt | Hardware confirmation before each session                         |
| Speech (Cognitive)   | Speaks letters, words, and numbers aloud on the beat              |
| Haptic Intensity     | Slider for device vibration strength (mobile only)                |

SETUP

## Calibration

### Eye Tracking Calibration (WebGazer)

Optional toggle in Oculomotor module. Uses front camera to estimate where the patient's eyes are pointing. Screening-level only — not clinical-grade eye movement recording.

**Requirements:** Chrome on Windows, Mac, or Android. Not supported on iPhone, iPad, or Safari.

**How to calibrate:** Enable toggle → tap Play → 9-point calibration grid → look at each dot and tap it → tap Start Session. Gaze crosshair appears on canvas during session.

**Accuracy note**  
WebGazer accuracy is approximately ±60–80 pixels depending on device and lighting. Screening-level — enough to detect clear trailing or anticipation, but not precise enough to replace a behavioral optometry evaluation. SOAP note includes appropriate disclosure automatically.

### Camera Positioning (Motion & Active Games)

Positioning guide appears in the setup overlay for each sub-mode. Start button enables automatically when required body landmarks are detected. Full-body modes wait until both ankles are visible.

| Mode                | Camera Shows                     | Distance |
|---------------------|----------------------------------|----------|
| Clapping            | Waist up — both hands visible    | 3–4 feet |
| Knee Taps / Midline | Waist to head, knees visible     | 3–5 feet |
| Marching            | Full body — head to feet         | 4–6 feet |
| Jumps               | Full body + clearance above head | 5–7 feet |
| All Active Games    | Full body                        | 4–7 feet |

**Camera zoom tip**  
A zoom slider (1×–2×) appears in the camera setup screen. Slide right so the patient can stand closer. Pose detection remains accurate up to approximately 1.5×.

EQUIPMENT

## Hardware Guide

EvoSensory works on any device with a browser — no accessories required. On a computer tap with **spacebar** or **Enter**. On a phone or tablet, tap anywhere on screen.

### **Wired headphones — required for spatial audio feedback**

The spatial audio is the core clinical mechanism: left ear tone = tapping early, right ear tone = tapping late. Without wired headphones this feedback is not available. Bluetooth adds 100–200ms of lag that defeats the feedback loop entirely. Any wired earbuds work — including old phone earbuds.

## Recommended Accessories

### **Vibrating Cushion — for added sensory feedback**

Place behind the patient (against the seatback) or under them on the seat. In Settings → XAC Pillow Rumble, choose when it fires:

#### **Correct tap (reward)**

Vibrates on clean hits. Best for early learners and sensory-seeking patients.

#### **Timing error (correction)**

Vibrates when patient taps early or late. Pairs with spatial audio cue.

### **External Button / Switch — for foot taps**

Any USB switch or jelly bean button works as foot tap input. Level 6 is designed for bilateral foot training. The app automatically sets a lower starting BPM for foot modality.

### **Xbox Adaptive Controller (XAC) — connects everything**

To connect external buttons/switches *and* a vibrating cushion in one device, the XAC is the recommended hub. AAC switches plug into the 3.5mm back jacks — no adapter needed. Button 0 = LEFT tap, Button 1 = RIGHT tap. USB connection means no Bluetooth lag.

The XAC gives you simultaneous L/R switch input for bilateral training and drives the vibration cushion — all in one device with a single USB connection.

## At a Glance

| Setup                     | Cost     | What It Adds                                                                                     |
|---------------------------|----------|--------------------------------------------------------------------------------------------------|
| Any device + browser      | \$0      | Full timing training, all modules, SOAP notes. Tap with spacebar, Enter, or screen tap.          |
| + Wired headphones        | ~\$10    | <b>Required for spatial audio.</b> Left ear tone = tapping early, right ear tone = tapping late. |
| + Vibrating cushion       | ~\$30–80 | Whole-body proprioceptive feedback — reward or error correction (your choice)                    |
| + USB switch / jelly bean | ~\$20–60 | External button for hand or foot input; larger tap target                                        |

| Setup                      | Cost   | What It Adds                                                                            |
|----------------------------|--------|-----------------------------------------------------------------------------------------|
| + Xbox Adaptive Controller | ~\$100 | Connects switches + cushion together. Bilateral L/R input + cushion output via one USB. |
| Chrome on computer         | —      | Required for Motion, Active Games, and Oculomotor eye tracking                          |

Evolving Therapy & Wellness

Vanessa Dorado Walker, MS, OTR/L · [vanessa@evolve-ot.com](mailto:vanessa@evolve-ot.com) · [www.evolve-ot.com](http://www.evolve-ot.com)

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